

Embeddings

You’ve learned that AI turns text into token IDs. But that’s just a number. It doesn’t say anything about what it means.

It’s the same as the number assigned to your Student ID. It might let you in the building, but it doesn’t tell anyone whether you are funny, into hockey, or the person who steals fries at lunch.



AI has a way to turn those token IDs into meaning. It’s called **Embedding**.

To see how this works, you and your friends decide to rate Coke and a cup of coffee on six characteristics: Sweet, Bitter, Fizz, Heat, Caffeine, and Dark.

Here are your results. We added a column for Token ID, even though that wasn’t part of your test.

Taste Profile - Coke vs. Coffee							
TOKEN	TOKEN ID	DIMENSIONS					
		SWEET	BITTER	FIZZ	HEAT	CAFFEINE	DARK
Coke	24317	9	1	10	2	3	8
Coffee	51820	1	9	0	9	8	10

If someone asked you, “Which drink has sweet of 9, bitter of 1, and fizz of 10?” you’d immediately answer Coke.

By rating each drink on those dimensions, you’ve done what AI does on a massive scale: you’ve represented a word with a row of numbers.

A row of numbers in a specific order, like this, is called a **Vector**. Each slot (Sweet, Bitter, etc.) is a **Dimension**. And each number is a **Value**.

WHAT ABOUT PEPSI?

Now add a third drink to the taste test: Pepsi. Score it on the same six dimensions and a problem shows up. Pepsi looks almost exactly like Coke: both sweet, both fizzy, both lightly caffeinated, neither bitter nor hot. On these six numbers alone, you cannot tell them apart.

Taste Profile - Coke vs. Pepsi vs. Coffee								
TOKEN	TOKEN ID	DIMENSIONS						
		SWEET	BITTER	FIZZ	HEAT	CAFFEINE	DARK	CITRUS
Coke	24317	9	1	10	2	3	8	1
Pepsi	38106	9	1	10	2	3	8	10
Coffee	51820	1	9	0	9	8	10	0

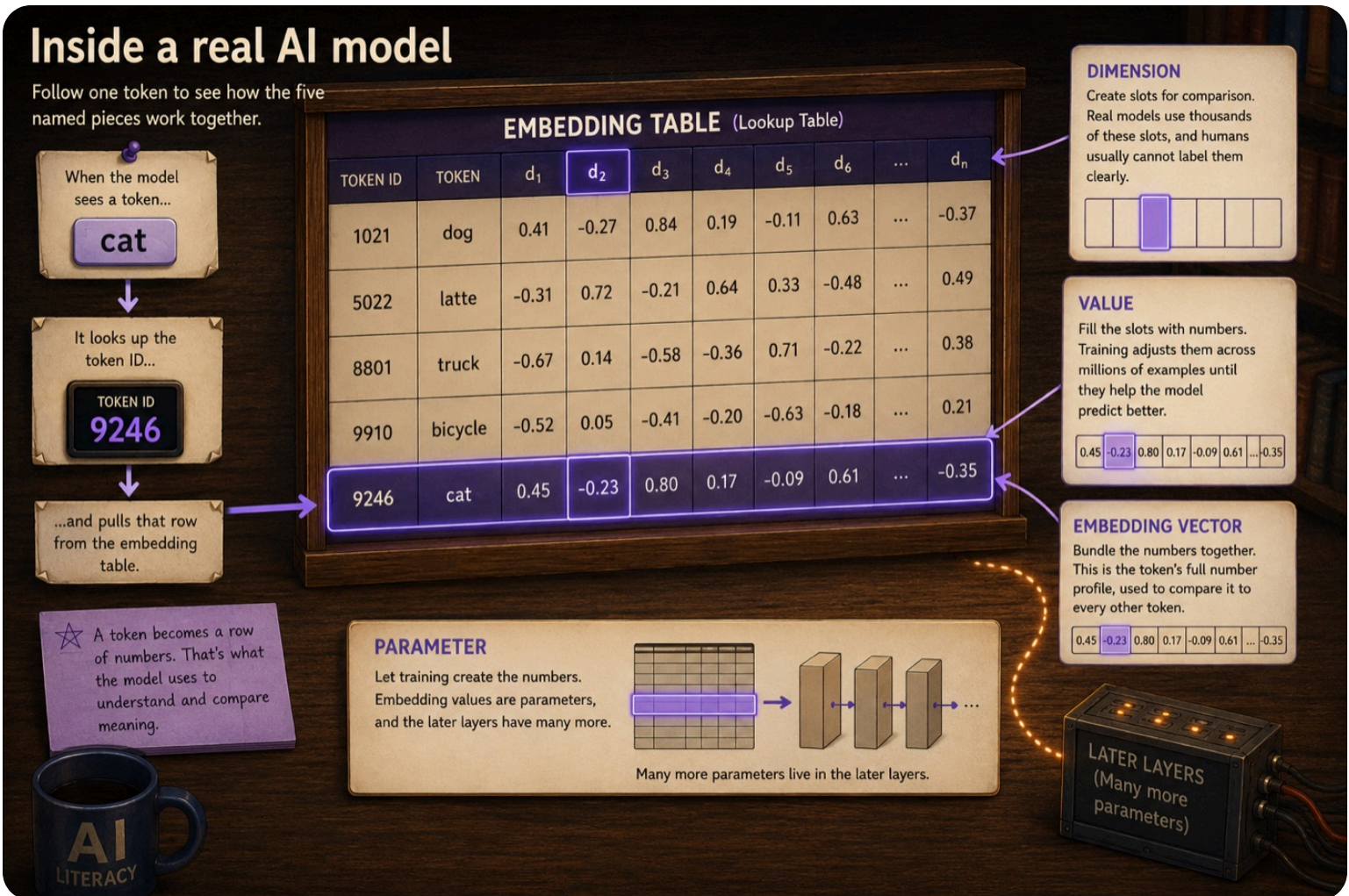
To separate them, you added a new dimension, **Citrus**. Pepsi scores high on it while Coke sits near zero, and only then do the two rows finally differ.

But AI can’t add a unique dimension like Citrus for every distinction between all the words in a language. So instead of a handful of labeled dimensions, real models give each token **thousands of dimensions**, learned during training.

Every token uses those same dimensions. Only the values change.

INSIDE A REAL MODEL

Here’s how it all fits together. The word **cat** is tokenized and assigned the token ID of 9246, which looks up a row in the **embedding table** (a giant lookup of every token’s numbers). That row is cat’s **embedding vector**: its full profile of values across the dimensions. The numbers that fill the table, plus many more throughout the model, are called **parameters**. Notice the dimensions aren’t named: the model decides what each one tracks, and we usually can’t tell.



DOES EVERY TOKEN GET ITS OWN VECTOR?

Yes. Remember how ‘unbelievable’ became three tokens? Each one (‘un’, ‘believ’, ‘able’) gets its own vector with thousands of values. The model does not start with the meaning of the full word. Combining those pieces into the meaning of the whole word happens later, in the layers.